Band-1 Hadamard Order 4804 — Miyamoto (Double Paley-I seed H_1200); in SPK via Paley-II (7^4); Gap 2052 Still OPEN

H_600/H_1200/C_1202/H_4804 PASS; 4804 in SPK via Paley-II prime power (q=2401=7^4); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_4612_Miyamoto`; `2026-09-15_Band1_Hadamard_4516_Miyamoto` (Double Paley-I pattern).
Lab. `/workspace/experiments/band1/hadamard_4804_miyamoto/`.
Cutoff. 2026-09-15T17:35:03Z (~1:35PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 600, 1200 and 4804 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1200} (H_{600} = Paley-I($q = 599$); H_{1200} = Sylvester-double(H_{600})). Paley-II path for H_{1200} is **FAIL** ($q = 599 \equiv 3 \pmod 4$; needs $\equiv 1$). Paley-I direct for H_{1200} is **SKIP** ($q = 1199 = 11 \cdot 109$ not prime). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 4804 **enters SPK** via classical Paley-II over the prime power $q = 2401 = 7^4 \equiv 1 \pmod 4$ ($n = 2(q + 1) = 4804$). On-box prime-only Paley-II constructor **SKIP** (2401 not prime; GF(7^4) not implemented). On-box verification used **Miyamoto** as an independent finite path. **Not** a Miyamoto-only bonus. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 4612 on box. This increment ships **order 4804** ($4804 = 4 \cdot 1201$, 1201 prime $\equiv 1 \pmod 4$) via Miyamoto with a **Double(Paley-I) seed** H_{1200}. Symmetric Paley conference C_{1202} over prime 1201; $H_{4804} = \text{Miyamoto}(H_{1200}, C_{1202})$. Unit tests **51/51 PASS**. SPK next gap remains **2052** (OPEN). Order 4804 **is in SPK** via Paley-II prime power ($q = 2401 = 7^4$). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 4804; Miyamoto; Paley type I; Double Paley-I; Paley type II prime power; Paley conference; constructive gap 2052; SPK; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in ± 1 satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{600} constructed and verified (Paley type I, $q = 599$). 2. H_{1200} constructed and verified (Sylvester double of H_{600}). Paley-II path for H_{1200} **FAIL** ($q = 599 \equiv 3 \pmod 4$). Paley-I direct **SKIP** ($q = 1199 = 11 \cdot 109$). 3. C_{1202} constructed and verified (Paley conference, $q = 1201$): $CC^\top = 1201I$, zero diagonal. 4. H_{4804} constructed and verified (Miyamoto). **Not** the Hadamard conjecture. 5. SPK: $4804 \in$ SPK via Paley-II prime power ($q = 2401 = 7^4$). Paley-I / Kronecker paths for 4804 **rejected**. Next constructive gap remains **2052** (OPEN). Verified H_{600}/H_{1200} enter SPK as Double Paley-I seeds.

2. Arithmetic and eligibility

Quantity	Value
Target order	$4804 = 4 \cdot 1201$
$q = 1201$	prime, $1201 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{1200}
Needed conference	order $1202 = q + 1$
H_{1200} path (used)	Double(Paley-I): $H_{600} = \text{Paley-I}(q = 599)$; $H_{1200} = \text{Sylvester-double}(H_{600})$
H_{1200} Paley-II	**FAIL**: $2(q + 1) = 1200 \Rightarrow q = 599 \equiv 3 \pmod 4$ (needs $\equiv 1$)
H_{1200} Paley-I direct	**SKIP**: $q = 1199 = 11 \cdot 109$ not prime
Conference path	Symmetric Paley over prime 1201
Paley-I for 4804	**REJECTED**: $q' = 4803 = 3 \cdot 1601$ not prime power
Paley-II for 4804	**PASS (SPK)**: $q' = 2401 = 7^4 \equiv 1 \pmod 4 \rightarrow n = 2(q' + 1) = 4804$; on-box prime-only constructor SKIP

Quantity	Value
Kronecker for 4804	**REJECTED**: only 4×1201; $1201 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{600}	Paley type I, $q = 599$	0	0.2913	**PASS**
H_{1200}	Sylvester double of H_{600}	0	0.783	**PASS**
H_{1200} Paley-II	$q = 599 \equiv 3 \pmod 4$	—	—	**FAIL** (needs $\equiv 1$)
H_{1200} Paley-I direct	$q = 1199 = 11 \cdot 109$	—	—	**SKIP** (not prime)
C_{1202}	Paley conference, $q = 1201$	0	1.4408	**PASS**
H_{4804}	Miyamoto(H_{1200}, C_{1202})	0	49.1629	**PASS**

Primary pipeline wall (incl. unit tests + SPK refresh) ≈ 53.1922 s. Compressed ± 1 matrices saved as `H600_pm1.npz`, `H1200_pm1.npz`, `C1202_pm1.npz`, `H4804_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior classical/SPK ships (through 4036/3908/3748/… + **1020/1116** Paley-II + **1032/1092/1152** Paley-I + **524/1048/548/1096/564/1128** Double Paley-I; **exclude** 4612/4516/4468/4388/4372/4196/4132/4084/3716/3524/3284/3076/3044/2836/2692/2564 Miyamoto-only) $\cup$ verified H_{600}/H_{1200} (Double Paley-I) $\cup$ Sylvester $\cup$ Paley I/II (incl. prime powers), closed under Kronecker (limit 4900). Order 4804 enters SPK via Paley-II prime power (7^4); on-box construction used Miyamoto independently.

Quantity	Value
Known count	1031
4804 in SPK	**True** (Paley-II prime power 7^4)
1200 in SPK	**True** (Double Paley-I seed)
600 in SPK	**True** (Paley-I seed)

Quantity	Value
1152 in SPK	**True** (prior Paley-I)
Next constructive gap	**2052** (OPEN)

5. Gap 2052 (method triage / non-finding ONLY)

Paley-I, Paley-II, Kronecker, and Miyamoto-with-Paley-conference paths for order 2052 all **FAIL** on-box arithmetic. This is a **non-finding**, not a counterexample. Gap **2052 remains OPEN**. Conjecture **OPEN**.

6. Non-claims

- No full Hadamard conjecture claim.
- `clay_closed = false` forever in this ship.
- GATE-04 collection **LOCKED**; no invented J.
- No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / finished-ToE claims.
- Order 4804 is **in SPK** via Paley-II prime power (not a Miyamoto-only bonus); on-box path was Miyamoto.

7. Next optional

- q=1217 → n=4868 via Double(PaleyI half=608) / Kronecker(4×304)
- q=1229 → n=4916 via PaleyII(q=613)
- method-attack SPK gap **2052**
- Track B privacy / Object 1 meter await

8. Reproducibility

```
cd /workspace/experiments/band1/hadamard_4804_miyamoto
python3 verify_hadamard_4804.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, `*.npz`, this PAPER.